

Code-Layout, Readability and Reusability

Date:	3 October 2026	Team ID:	UNM1611SHARVESH RAM
Project Name:	AI BlogNest API	Maximum Marks:	5 Marks

Code Quality & Architecture Assessment:

This document evaluates the codebase in terms of its modular layout, readability, adhering to Node.js/Express clean code conventions, and high component reusability.

Code Layout Checklist:

S.No	Quality Parameter	Description	Followed	Implementation Remarks in AI BlogNest
1	Consistent Indentation	Uniform spacing and line breaks throughout code.	Yes	Strict 2-space indentation formatted with Prettier and ESLint standards.
2	Proper File Structure	Files and folders logically organized into modules.	Yes	Clear MVC directory: config/, controllers/, middleware/, models/, routes/, services/.
3	Meaningful Names	Variables reflect their intent and data type.	Yes	Descriptive identifiers: existingUser, resolvedCategory, cleanGeneratedText, callGemini.
4	Function / Method Names	Functions named using clear verb-noun patterns.	Yes	Explicit actions: register, login, profile, createBlog, generateBlog, summarizeBlog.
5	Code Comments	Inline comments explain non-obvious logic.	Yes	Documented Gemini SDK initialization, fallback model selection, and regex patterns.
6	Modular Design	Code split into single-responsibility units.	Yes	AI service logic completely decoupled from HTTP controllers and database models.
7	No Redundant Code	Duplicate or dead code is eliminated.	Yes	Shared helper functions for text sanitization, token signing, and category mapping.
8	Error Handling	Graceful exception catching and error forwarding.	Yes	All async controllers pass errors to centralized next(error) handling middleware.

Code-Layout, Readability and Reusability — Modules & Quality Score

Reusable Components and Modules:

S.No	Component / Module	Language / Tech	Where Reused	Reusability Level
1	authMiddleware.js	JavaScript / JWT	Guards all protected endpoints: /api/auth/profile, /api/blogs (POST, PUT, DELETE), /api/ai/generate-blog	High
2	errorMiddleware.js	JavaScript / Express	Global pipeline catcher for all unhandled async errors, validation issues, and MongoDB cast exceptions	High
3	geminiService.js	JS / Google SDK	Centralized GoogleGenerativeAI client; invoked by blog generator and summarization controllers	High
4	cleanGeneratedText	JS / RegExp	Utility function stripping markdown artifacts, hashes, and formatting across all AI text generators	High
5	inferCategory	JS / Keyword Map	Rule-based classifier mapping prompts into 9 distinct domains when user leaves category empty	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Evaluator Comments & Evidence
Code Layout & Structure	5 / 5	Flawless MVC separation. Clear segregation into controllers, routes, models, and services.
Readability & Style	5 / 5	Self-documenting variable and function names. Concise and clean code formatting.
Reusability	5 / 5	Highly reusable middleware for authentication, error catching, and centralized Gemini client.
Documentation & Comments	5 / 5	Comprehensive JSDoc comments, clean README.md setup guide, and postman collection documentation.
Overall Score	5.0 / 5.0	Exemplary codebase demonstrating professional backend and GenAI engineering practices.